Audience Sensitivity, Content Compliance & Format Preference

Analysis of the France Top 50 Playlist, 18 May 2024 – 27 Nov 2025

Prepared for Atlantic Recording Corporation (Unified Mentor project). Data: 27,800 raw daily playlist rows, 555 snapshots, 724 distinct tracks.

1. Executive overview

The brief assumes a French audience that is cautious about explicit lyrics and prefers structured albums over rapid-release singles. The playlist data only partly supports that picture. Four findings stand out:

- **Explicit content is the majority, not the exception.** 56.3% of chart slots are explicit, rising to 62.4% in the Top 10. Explicit tracks are slightly over-represented near the top (Content Acceptance Score 102).
- **But explicit tracks are less durable.** They score lower on popularity (73.7 vs 80.9) and stay on the chart for a median of 7 days against 13 for clean tracks.
- **Singles, not albums, hold the top of the chart.** Singles take 50.7% of Top 10 slots, score 80.8 popularity vs 73.3 for album tracks, and last about twice as long (mean 59 vs 27.7 days). Album tracks dominate only the lower half of the chart.
- **Bigger albums do not strengthen individual tracks.** The Album Size Impact Index is 96 (below 100). The effect is driven by clean catalogue and deluxe releases; for explicit (largely rap) albums, size makes no measurable difference.

2. Data and preparation

The file contains one row per chart position per day. The following validation steps were applied (all logged in the dashboard's Data quality tab):

Check	Result	Action
50 entries per day	554 of 555 days have exactly 50 rows	1 Mar 2025 held two different chart snapshots (100 rows; 19 rows were exact duplicates, 31 positions differed): first snapshot kept. 2 Feb 2025 has 49 rows after the corrupt row below is removed.
Corrupt rows	1 row (2 Feb 2025, rank 2) had no song title, zero duration and zero popularity	Removed
Missing calendar days	14 Mar, 25 Mar, 11 Jul, 13 Aug 2025 absent from the file	Left as gaps (no imputation)
Popularity = 0	72 rows	Treated as missing for popularity statistics only
Duration	Converted ms → minutes; buckets: short <2.5, medium 2.5–3.5, long >3.5 min	Applied
album_type labels	Lower-cased/trimmed: single, album, compilation (9 rows)	Compilations (Christmas/Halloween seasonals) kept but not analysed separately

Check	Result	Action
is_explicit consistency	5 tracks carry both TRUE and FALSE across days (e.g. different versions of the same title)	Flagged; row-level value used

Result: 27,749 clean rows, 724 unique tracks. Because the same tracks appear on many days, daily rows are not independent observations; where it matters, track-level views (e.g. days on chart) are used alongside slot-level views.

3. Explicit content sensitivity

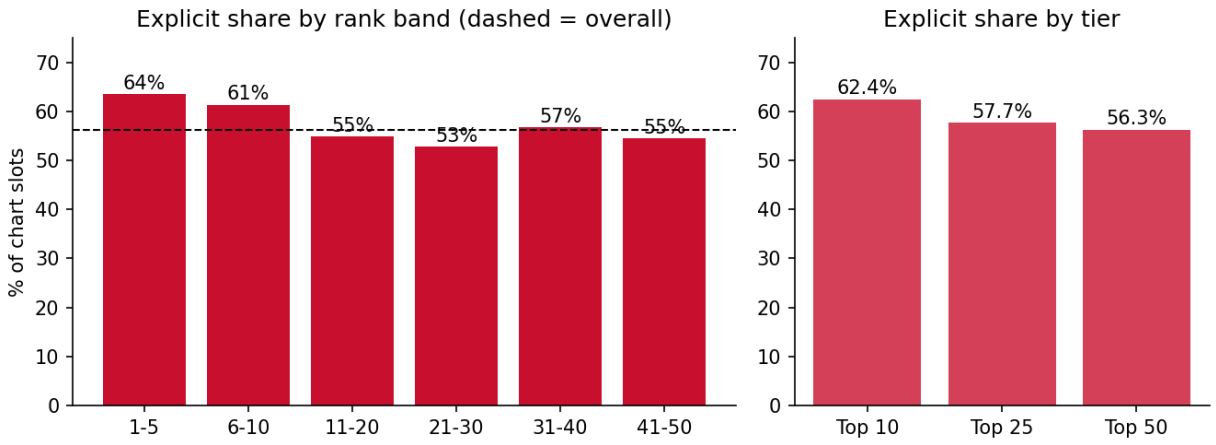


Figure 1. Explicit share by rank band and tier.

Explicit share declines with rank: 63.6% of positions 1–5 against about 52.8% in positions 21–30. The top of the French chart is therefore not filtering explicit material out; if anything, it favours it. Explicit tracks average rank 24.99 vs 26.16 for clean tracks (lower = better).

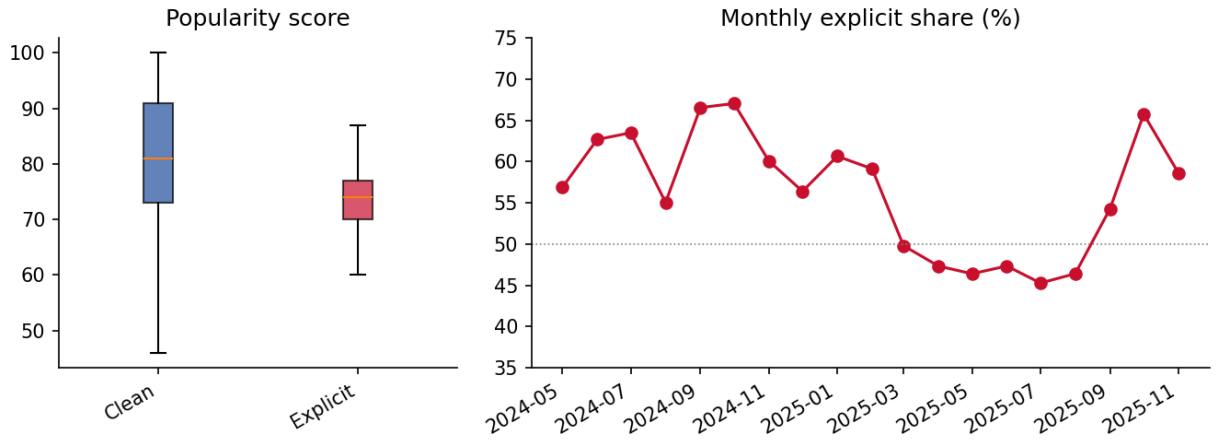


Figure 2. Popularity by content type and monthly explicit share.

Popularity points the other way. Clean tracks average 80.9 against 73.7 for explicit (Mann–Whitney $p < 0.001$). Reading both signals together: explicit tracks enter high and fast but burn out sooner (median 7 days on chart vs 13 for clean, 480 explicit vs 244 clean tracks), while clean tracks build slower and persist. Note that the popularity score is a rolling Atlantic API measure and is only weakly related to rank (Spearman $\rho = -0.18$), so it should not be read as a proxy for chart success.

Explicit share is also seasonal. It peaked near 67% in Oct 2024 and 66% in Oct 2025 (autumn rap album season) and fell to roughly 45–50% between March and August 2025, when clean pop and international singles filled more slots.

4. Release format preference

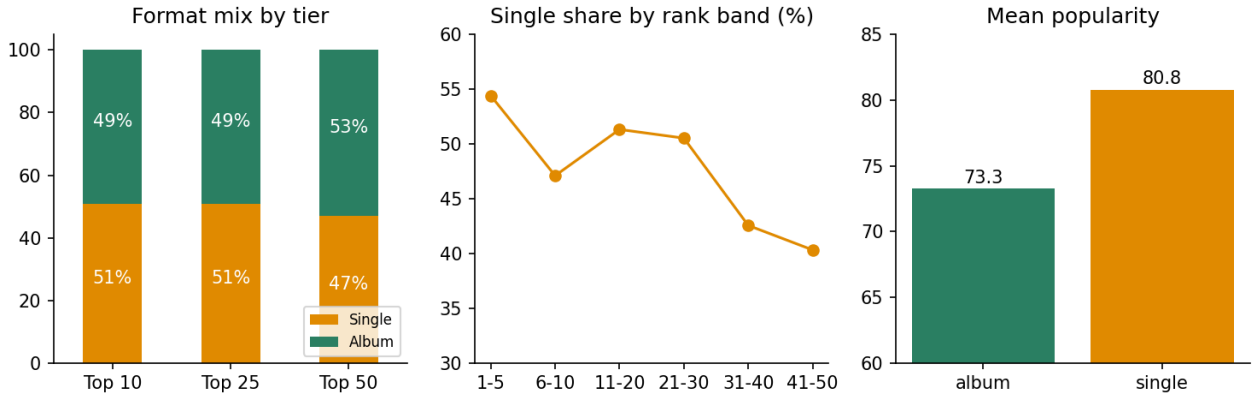


Figure 3. Format mix by tier, single share by rank band, and mean popularity.

Singles and album tracks are close to parity overall (47.1% / 52.9%), but the split changes with rank. Singles hold 50.7% of the Top 10 and 51% of the Top 25; album tracks take over below rank 25 (about 57% of positions 26–50). Singles average rank 24.12 vs 26.73 for albums and have higher popularity (80.8 vs 73.3).

This does not contradict the idea that France values album structure; it shows that album releases act as a breadth engine (469 distinct album tracks charted vs 250 singles) while singles provide the sustained, high-ranking core (mean 59 days on chart vs 27.7). Album tracks are also much more likely to be explicit (72.4% vs 38.2% for singles), so the format and content effects overlap.

5. Album structure impact

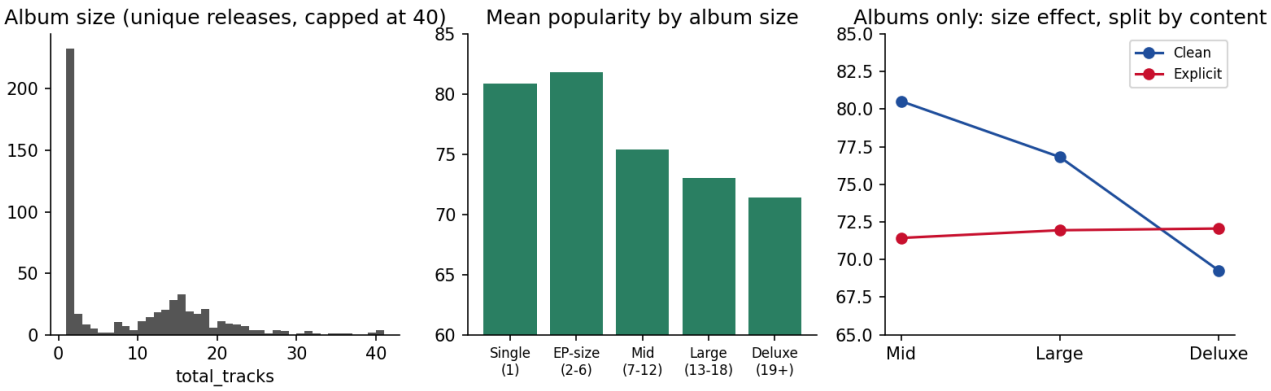


Figure 4. Album size distribution, popularity by size band, and the same effect split by content type.

Size band	Chart slots	Mean popularity	Mean rank	Explicit share
Single (1)	11,310	80.9	23.6	38%
EP-size (2-6)	1,304	81.8	28.2	20%
Mid (7-12)	4,064	75.4	25.4	62%
Large (13-18)	8,277	73.1	26.7	77%
Deluxe (19+)	2,794	71.4	28.4	77%

Popularity falls steadily as albums get larger, from about 81.8 for EP-size releases to 71.4 for 19+ track deluxe editions (Album Size Impact Index 96). A naive reading says large albums dilute their tracks. The split in Figure 4 shows the story is narrower: among clean album tracks the size effect is clear (ρ = -0.258; popularity ~80 at 7–12 tracks to ~69 at 19+), whereas among explicit album tracks

it is nil (ρ = -0.002; popularity ~72 at every size). Large clean albums are mostly older catalogue and deluxe/reissue content, which carries lower popularity per track.

Dilution vs concentration. When an album places several tracks in the chart on the same day, per-track popularity falls (74.2 for one track, 66.5 when five or more chart together) but average rank improves (27.7 → 23.8). Album "flooding" therefore concentrates chart real estate while spreading attention across tracks. It occurs on only about 1.4% of album-days, but on those days it can occupy 5–18 slots.

6. Song duration

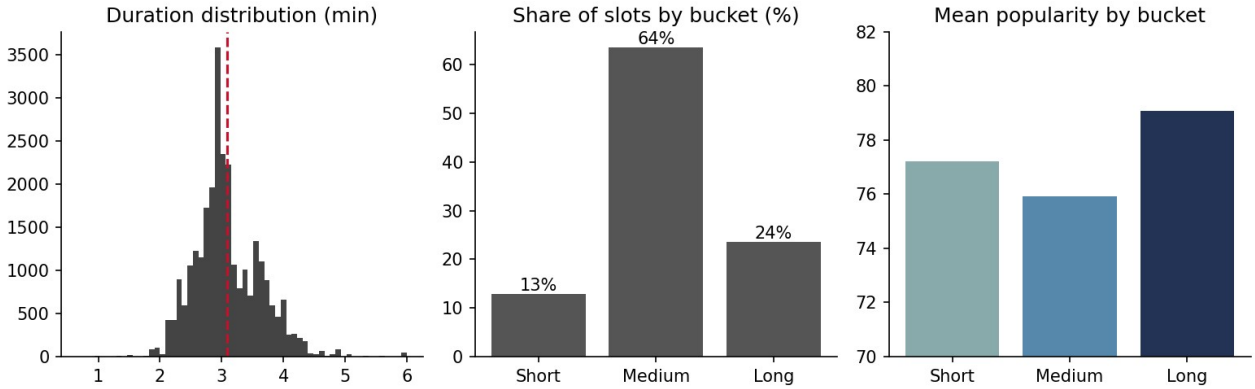


Figure 5. Duration distribution, bucket shares and mean popularity.

Duration is tightly clustered: mean 3.09 min, median 3.00 min, interquartile range 2.76–3.46 min. About 64% of slots are medium (2.5–3.5 min), 24% long and 13% short. Duration has no relationship with rank (ρ = 0.008, not significant) and only a very small link to popularity (ρ = 0.10). Long tracks look slightly more popular (79.1) but 66% of them are explicit, so this largely reflects rap-album cuts, not a general preference for length. French listeners do not penalise either short or long tracks; the 3-minute norm is a convention, not a ranking driver.

7. Preferred content profiles

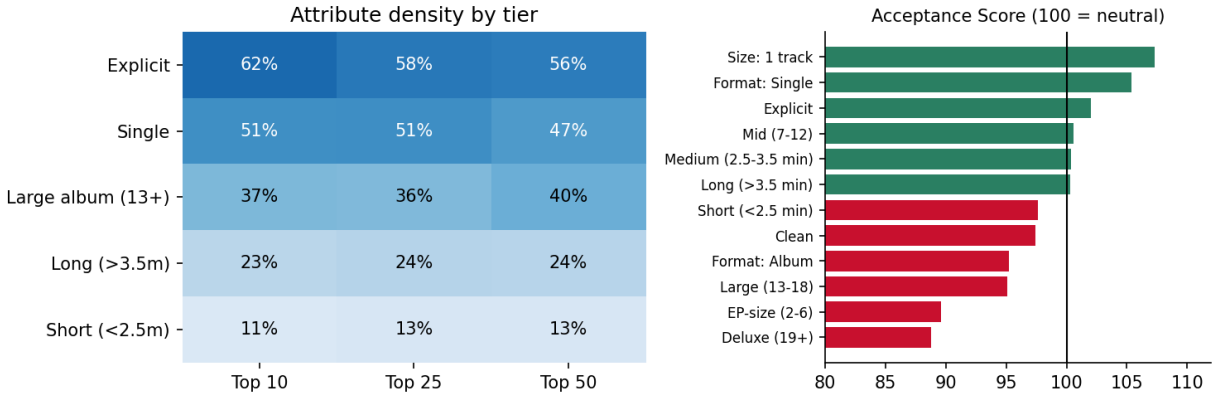


Figure 6. Attribute density by tier and Content Acceptance Score.

Profile (content / format / length)	Top 10	Top 25	Top 50
Explicit / Album / Medium	28.5%	25.3%	24.7%
Clean / Single / Medium	21.5%	22.1%	21.7%
Explicit / Album / Long	7.1%	9.2%	10.2%
Explicit / Single / Medium	8.8%	9.8%	9.1%
Clean / Album / Medium	0%	0%	8.1%

Two archetypes account for roughly half of the chart: the explicit album cut of medium length (24.7% of Top 50 slots) and the clean medium-length single (21.7%). At the very top the explicit long single becomes prominent (12% of the Top 10). Using the Content Acceptance Score (rank-weighted share divided by raw share, 100 = neutral), the segments that over-index near the top are singles (105), explicit tracks (102) and single-track releases (107); the segments that under-index are deluxe albums (89), EP-size releases (90), album tracks generally (95) and clean tracks (97).

8. KPI summary

KPI	Top 10	Top 25	Top 50
Explicit Content Share (%)	62.44	57.70	56.27
Clean Content Dominance Ratio (clean:explicit)	0.60	0.73	0.78
Single vs Album Track Ratio (single:album)	1.03	1.04	0.89
Average Song Duration (min)	3.09	3.08	3.09
Album Size Impact Index	99.65	98.42	96.44
Content Acceptance Score - Explicit	100.00	101.17	102.00
Content Acceptance Score - Clean	100.00	98.41	97.43

Definitions: Clean Content Dominance Ratio = clean slots ÷ explicit slots (below 1 means explicit dominates). Single vs Album Ratio = single slots ÷ album slots. Album Size Impact Index = mean popularity of 13+ track albums ÷ mean popularity of 2–12 track albums × 100. Content Acceptance Score = rank-weighted share (weight = (51 − rank)/50) ÷ unweighted share × 100.

9. Recommendations

These follow from the patterns above; they are hypotheses to test, not guarantees.

- **Explicit-content risk.** Treat explicit content as commercially normal in France’s Top 50, but plan for shorter shelf life. For explicit lead releases, prepare a clean/radio edit to extend reach and longevity, and time them against the autumn peak in explicit share.
- **Release format.** Lead campaigns with singles for Top 10 positioning, then use the album to add breadth in the lower half of the chart. A single-first rollout matches how the chart actually behaves.
- **Album size.** For clean/pop projects, keep standard editions to roughly 7–12 tracks and avoid padding with deluxe tracks; for rap albums, size is not a penalty, but concentrate promotion on 2–3 focus tracks to avoid per-track dilution.
- **Duration.** Do not edit tracks to chase a length target. Stay near the 2.5–3.5 minute norm for pop singles, with no evidence of penalty for longer rap tracks.
- **Playlist pitching & localisation.** Pitch clean singles for steady, long-running placement and explicit album cuts for peak-season and top-rank bursts. Localisation of global releases should prioritise a French-market single and a clean version rather than a re-sized album.

10. Limitations

- Observational data: associations are not causal, and the popularity score is a rolling API metric, not streams.
- The Spotify-style explicit flag is a labelling attribute, not a legal or regulatory compliance judgement.

- Artist concentration is notable (Werenoi alone is 7.9% of slots; top 5 artists 19%), so a few catalogues influence the results.
- Four missing days and one duplicated day were handled as described; the dataset has no stream counts, genre or language fields, so “French-language” vs international content could not be separated.
- Daily rows repeat the same tracks, so slot-level p-values overstate precision; effect sizes and track-level checks are more reliable guides.

Appendix: deliverables

Streamlit dashboard (app.py) with date range, rank-tier, explicit and album-type filters; shared logic in fr_top50.py; reproducible analysis in run_eda.py; cleaned dataset Atlantic_France_clean.csv.